

EXPERIMENT NO. : 5

DATE: 10/09/2026

1. AIM

To perform and verify Elmore delay analysis for a given RC lumped model and determine the delay at different nodes for different points of interest.

2. TOOL

Cadence Virtuoso

3. THEORY

The Elmore delay is a first-order approximation used to estimate the propagation delay of an RC network. For an RC ladder, the delay at a particular node depends on the resistance along the path to that node and the capacitances downstream of each resistance.

The Elmore delay at a node is calculated using $t_E = \sum_k R_i C_{downstream,i}$, where R_i is the resistance in the path to the node and $C_{downstream,i}$ is the total capacitance connected downstream of that resistance.

The given circuit consists of five resistors and five capacitors forming an RC lumped network, with five points of interest P1 to P5. The transient response is simulated in Cadence Virtuoso, and the time taken by each point to reach 10%, 50%, and 90% of its final voltage is measured.

The simulated delay values are compared with the corresponding Elmore delay calculated for each point to verify the accuracy of the Elmore delay approximation.

4. TABLES

Table 4.1: Design Parameters

Parameter	Value	Unit
V_{dd}	5	V
R1	10	Ω
R2	12	Ω
R3	14	Ω
R4	16	Ω
R5	18	Ω
C1	12	pF
C2	14	pF
C3	16	pF
C4	18	pF
C5	10	pF

Table 4.2: Result comparison between simulation and theoretical values at 50% of V_{DD}

Point	Elmore RC	Calculated Delay(0.7*RC)	Simulated Delay	Error	Unit
P1	700	490	220.58	-54.9837%	ps
P2	1396	977.2	441.22	-54.8485%	ps
P3	2012	1408.4	891.4	-36.7083%	ps
P4	2460	1722	1213.64	-29.5215%	ps
P5	2640	1848	1090.99	-40.9637%	ps

Table 4.3: Result comparison between simulation and theoretical values at 10% of V_{DD}

Point	Elmore RC	Calculated Delay(0.1*RC)	Simulated Delay	Error	Unit
P1	700	70	13.9	-80.1429%	ps
P2	1396	139.6	90.46	-35.2006%	ps
P3	2012	201.2	130.4	-35.1889%	ps
P4	2460	246	332.43	+35.1341%	ps
P5	2640	264	274.14	+3.8409%	ps

Table 4.4: Result comparison between simulation and theoretical values at 90% of V_{DD}

Point	Elmore RC	Calculated Delay(2.2*RC)	Simulated Delay	Error	Unit
P1	700	1540	2059.63	+33.7422%	ps
P2	1396	3071.2	2239.67	-27.0751%	ps
P3	2012	4426.4	3059.1	-30.8897%	ps
P4	2460	5412	3382.81	-37.4943%	ps
P5	2640	5808	3252.14	-44.0059%	ps

5. CALCULATION FOR TRANSMISSION GATE BASED D FLIP FLOP

$$C_{total} = C_1 + C_2 + C_3 + C_4 + C_5 = 12 + 14 + 16 + 18 + 10 = 70pF$$

$$t_{E,P1} = R_1(C_1 + C_2 + C_3 + C_4 + C_5)$$

$$= 10(12 + 14 + 16 + 18 + 10) = 10(70) = 700ps$$

$$t_{E,P2} = R_1(C_1 + C_2 + C_3 + C_4 + C_5) + R_2(C_2 + C_3 + C_4 + C_5)$$

$$= 10(70) + 12(58) = 700 + 696 = 1396ps$$

$$t_{E,P3} = R_1(C_1 + C_2 + C_3 + C_4 + C_5) + R_2(C_2 + C_3 + C_4 + C_5) + R_3(C_3 + C_4 + C_5)$$

$$= 10(70) + 12(58) + 14(44) = 700 + 696 + 616 = 2012ps$$

$$t_{E,P4} = R_1(C_1 + C_2 + C_3 + C_4 + C_5) + R_2(C_2 + C_3 + C_4 + C_5) + R_3(C_3 + C_4 + C_5) + R_4(C_4 + C_5)$$

$$= 10(70) + 12(58) + 14(44) + 16(28) = 700 + 696 + 616 + 448 = 2460ps$$

$$t_{E,P5} = R_1(C_1 + C_2 + C_3 + C_4 + C_5) + R_2(C_2 + C_3 + C_4 + C_5) + R_3(C_3 + C_4 + C_5) + R_4(C_4 + C_5) + R_5C_5$$

$$= 10(70) + 12(58) + 14(44) + 16(28) + 18(10)$$

$$= 700 + 696 + 616 + 448 + 180 = 2640ps$$

6. CIRCUIT DIAGRAMS & WAVEFORMS

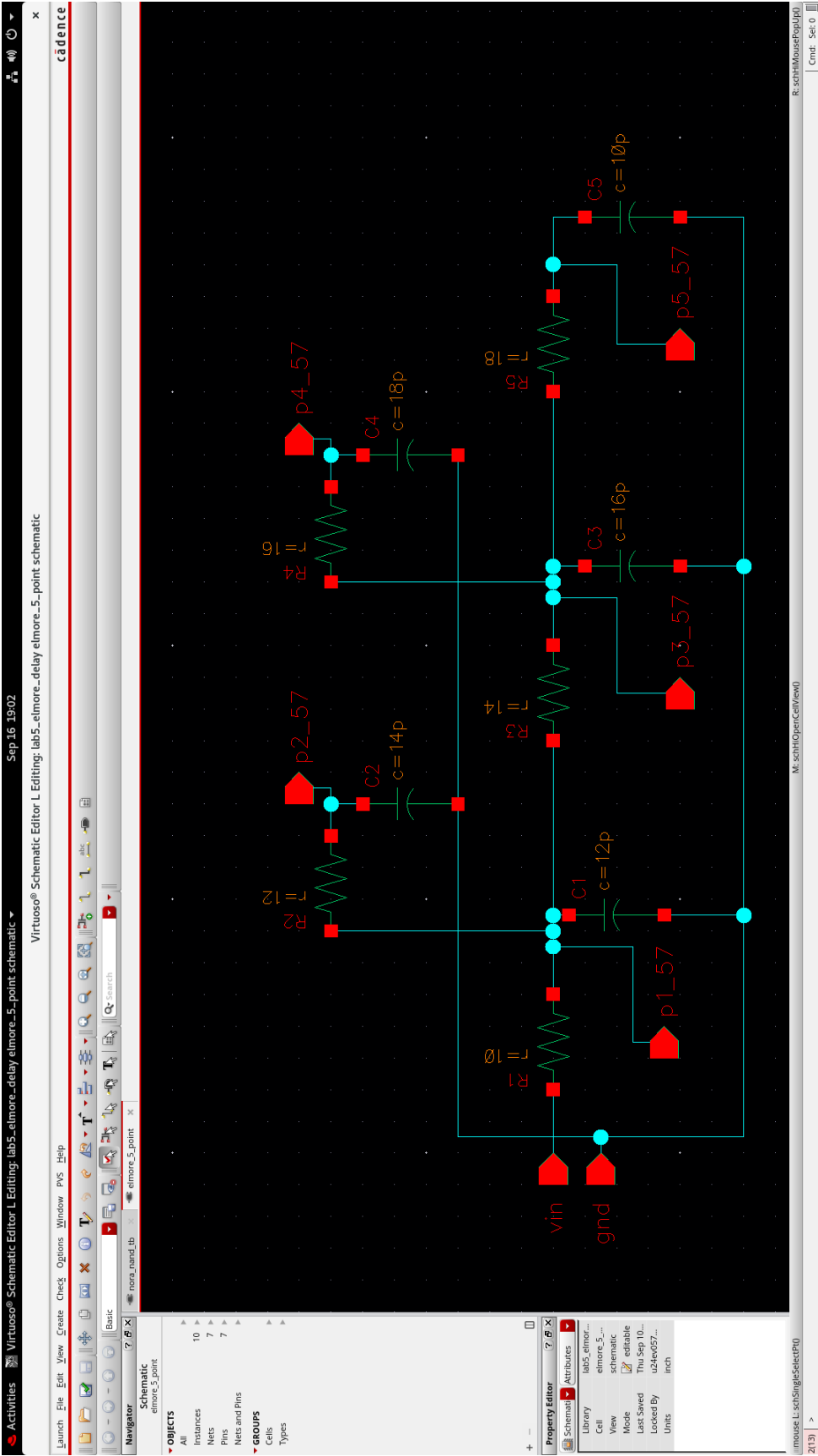


Figure 6.1: Schematic

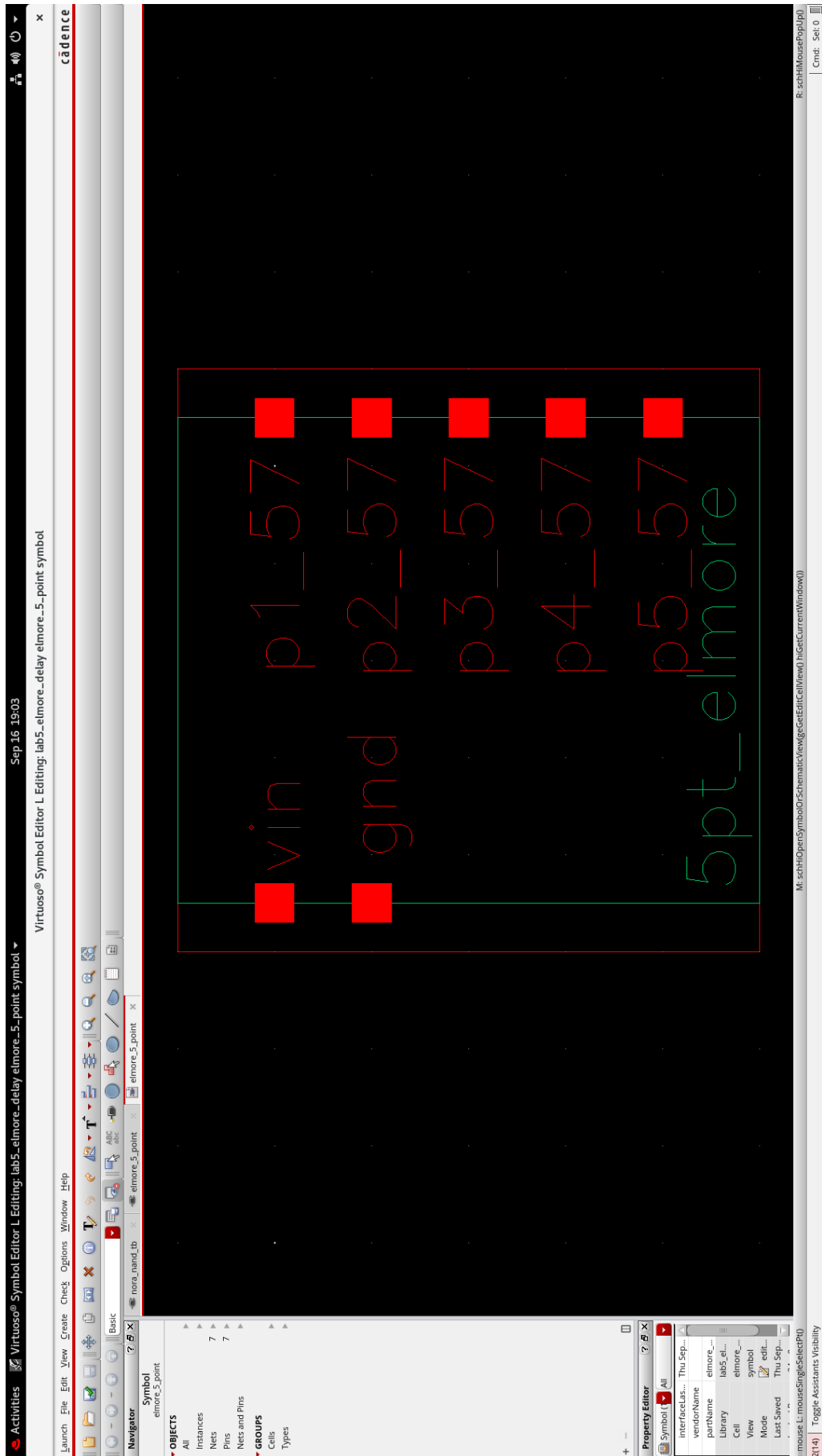


Figure 6.2: Symbol

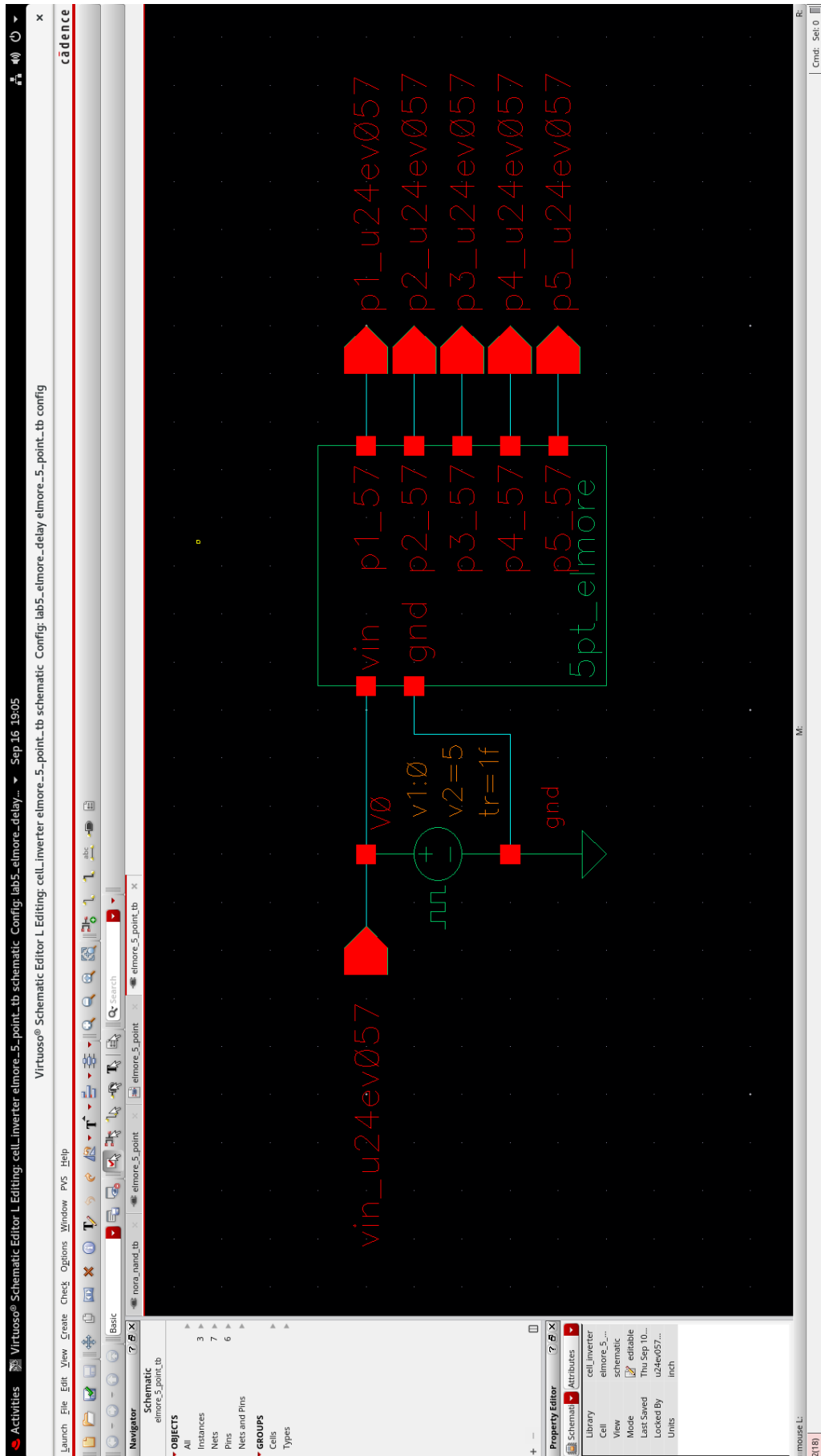


Figure 6.3: Tb

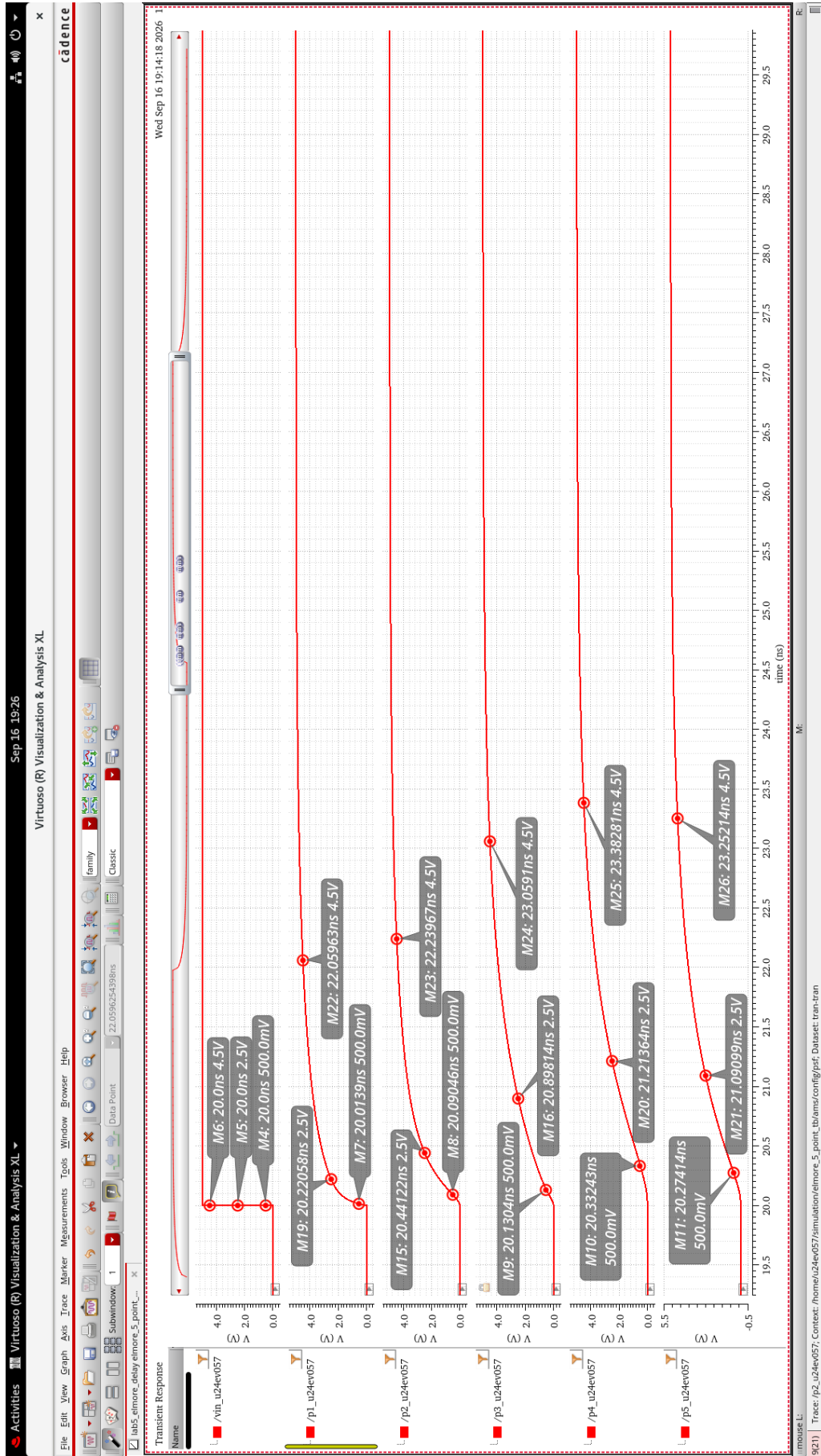


Figure 6.4: Tran

7. RESULT

The Elmore delay analysis of the given five-stage RC lumped network was successfully performed and verified. The theoretical delays at 10%, 50%, and 90% of V_{DD} were compared with the corresponding simulated delays at points P1 to P5, demonstrating the variation of propagation delay along the RC network.